



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

9189/1

PAPER 1 Theory

JUNE 2015 SESSION

2 hours

Additional materials:

Answer paper

Data Booklet

Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer six questions.

Answer two questions from Section A, one question from Section B, two questions from Section C and one other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 8 printed pages.

Copyright: Zimbabwe School Examination Council, J2015.

Section A

Answer at least two questions from this section.

- 1 (a) (i) Define the phrase *first ionization energy* of an element.
- (ii) Write an equation for the first ionization energy of chlorine. [2]
- (b) (i) Table 1 shows the first ionisation energies of four elements.

Table 1

element	1 st ionization energy/kJ mol ⁻¹
magnesium	736
aluminium	577
phosphorus	1 060
sulphur	1 000

Explain why

- magnesium has a higher first ionisation energy than aluminium,
- sulphur has a lower first ionisation energy than phosphorus.

[4]

- (ii) The successive ionization energies of an element, M, in kJ mol⁻¹ are:

1 st	2 nd	3 rd	4 th	5 th
740	1 500	7 700	10 500	13 600

State, with reasons, the group of the periodic table to which M belongs. [3]

- (e) State and explain the variation of electrical conductivity of Period (III) elements, Na to Ar. [3]

[Total:12]

- (a) Fig. 2.1 shows the structural formulae of isomeric hydrocarbons, P and Q.

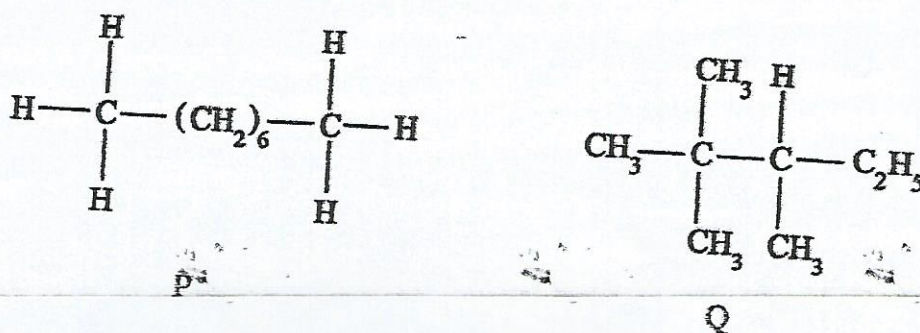


Fig. 2.1

- (i) Identify, with reasons, the hydrocarbon with a higher boiling point.
- (ii) Calculate the percentage of carbon by mass in 1 mole of P.
- (b) Fig. 2.2 shows the trend in the boiling points of the hydrides of Group (VI) elements, oxygen to tellurium. [4]

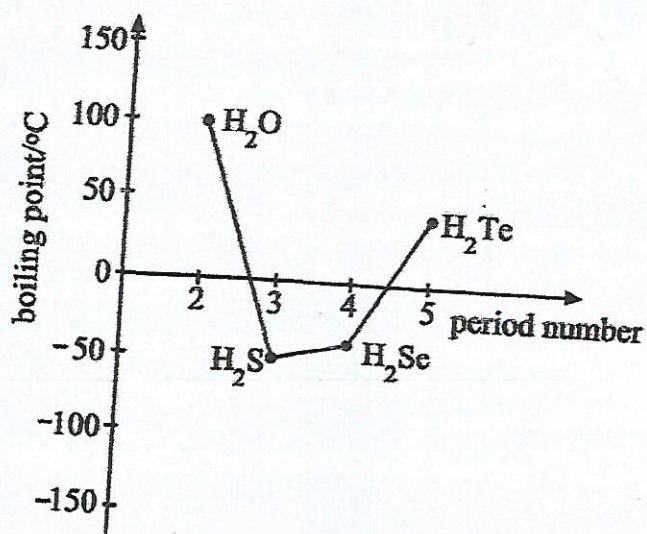


Fig. 2.2

Explain the trend in the boiling points of the hydrides.

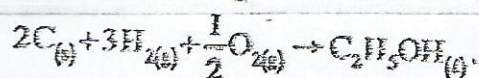
[5]

- (e) (i) Show, with the aid of a diagram, the bonding in boron trifluoride.
 (ii) Explain the reactivity of boron trifluoride.
 (iii) Draw a diagram to illustrate the shape of the product formed when boron trifluoride reacts with ammonia.

[3]

[Total: 12]

- 3 (a) The equation for the formation of ethanol is



- (i) Define the term *standard enthalpy change of formation*, ΔH_f^θ .
 (ii) Calculate the ΔH_f^θ of ethanol given that:

$$\Delta H_c^\theta(\text{C}) = -393.5 \text{ kJ mol}^{-1}$$

$$\Delta H_c^\theta(\text{H}_2) = -285.8 \text{ kJ mol}^{-1}$$

$$\Delta H_c^\theta(\text{C}_2\text{H}_5\text{OH}) = -1371 \text{ kJ mol}^{-1}$$

- (iii) Suggest why the ΔH_f^θ of ethanol cannot be measured directly.

[5]

- (b) Explain why the ΔH_f^θ of H_2O is the same as the ΔH_c^θ of H_2 .

[2]

- (c) In an experiment to measure the ΔH_c^θ of ethanol by calorimetry the following results were obtained:

volume of water in calorimeter	=	200.00 cm ³
mass of burner and ethanol at the beginning	=	81.00 g
mass of burner and ethanol at the end	=	77.78 g
initial temperature of water	=	24.0 °C
final temperature of water	=	89.0 °C

- (i) Calculate the

1. heat energy absorbed by the water,
2. enthalpy change of combustion, ΔH_c^θ , of ethanol.
 [Use of the data booklet is recommended]

- (ii) Suggest why the ΔH_c^θ value of ethanol given in a(ii) is different from the experimental value.

[5]

[Total: 12]

Section B

Answer at least one question from this section.

- 4 (a) Table 4.1 shows pH values of 1.00 mol dm^{-3} solution of Period (III) chlorides.

Table 4.1

chloride	pH
NaCl	7.0
MgCl_2	6.5
AlCl_3	3.0
SiCl_4	2.0
PCl_5	2.0

- (i) Write a chemical equation for the reaction of each chloride with water.
 (ii) Explain the trend in the pH values of the solutions.

[9]

- (b) State, with reasons, how the melting point of AlCl_3 compares to that of PCl_5 . [3]
 [Total: 12]

- 5 (a) (i) State with reasons, the physical states of chlorine, bromine and iodine at room temperature,
 (ii) explain the trend in the volatilities of the three elements in (i).

[5]

- (b) (i) Write a general equation for the decomposition of the nitrates of Group (II) elements.
 (ii) Explain the variation in the thermal stability of the nitrates of Group (II) elements.

[5]

- (c) Given that zinc nitrate decomposes like the nitrates of Group (II) elements, state two observations made when a sample of zinc nitrate is strongly heated. [2]
 [Total: 12]

Section C

Answer at least two questions from this section

- 6 (a) Fig. 6.1 shows the structure of cinnamaldehyde.

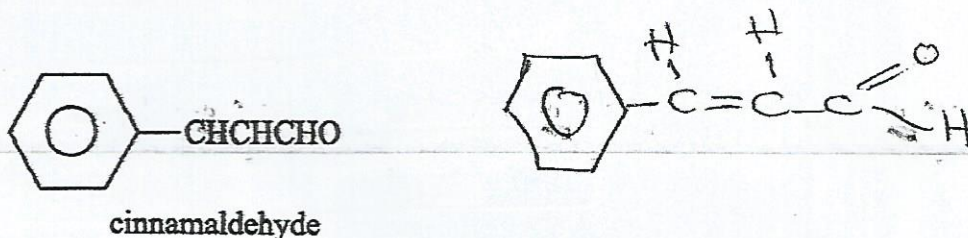


Fig. 6.1

- (i) Name
1. any two functional groups present in cinnamaldehyde,
 2. the type of isomerism that cinnamaldehyde exhibits.
- (ii) Draw structures to show the isomers of cinnamaldehyde. [4]
- (b) (i) Describe the reaction mechanism of cinnamaldehyde with HCN.
- (ii) Explain why the reaction in (b)(i) is carried out in the presence of aqueous NaOH.
- (iii) Give the structure of the organic product obtained when the compound produced in (b)(i) undergoes hydrolysis in aqueous NaOH. [6]
- (c) Describe a test giving the reagents, conditions and observations, to distinguish between benzaldehyde and cinnamaldehyde. [2]
- [Total:12]

- (a) Fig. 7.1 shows how benzene can be converted into other organic aromatic compounds.

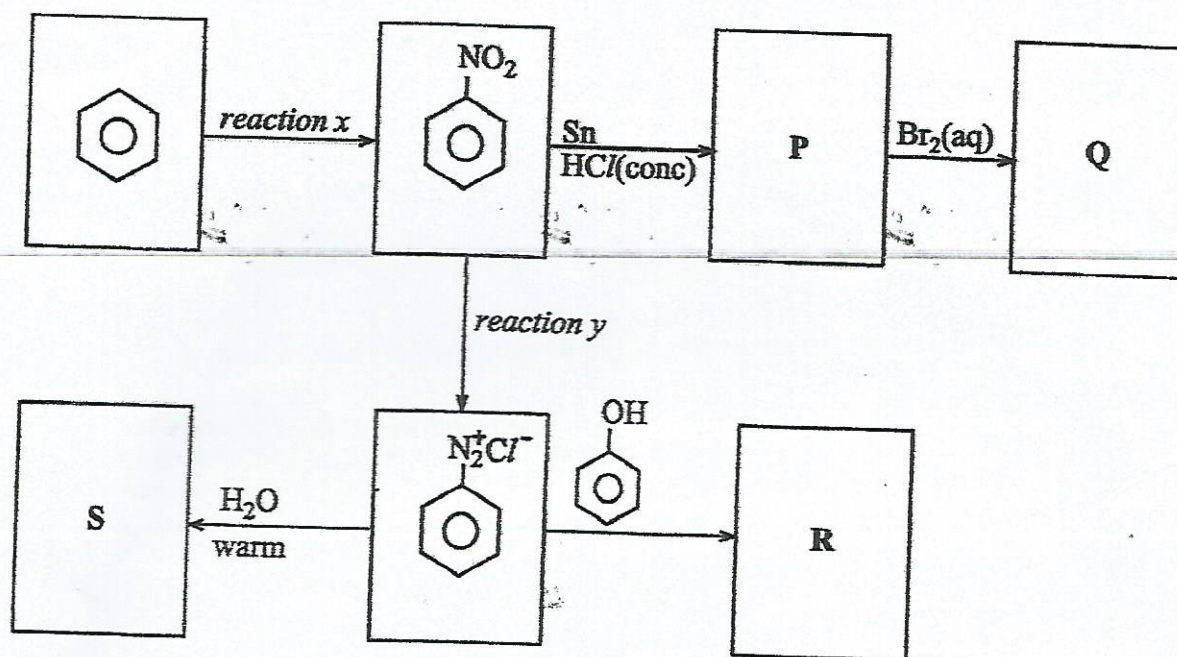


Fig. 7.1

- (i) Give reagents and conditions for
 1. *reaction x*,
 2. *reaction y*.
 - (ii) Draw the displayed structural formulae of compounds **P**, **Q**, **R** and **S**.
 - (iii) Name the type of reaction that forms **R**.
 - (iv) State one use of **R**.
- (b) State and explain what happens when
1. benzene diazonium chloride is placed in warm water,
 2. ethylamine, dissolved in hydrochloric acid, is evaporated.

[8]

[4]

[Total:12]

- 8 (a) Fig. 8.1 shows the formula of a synthetic polymer X.

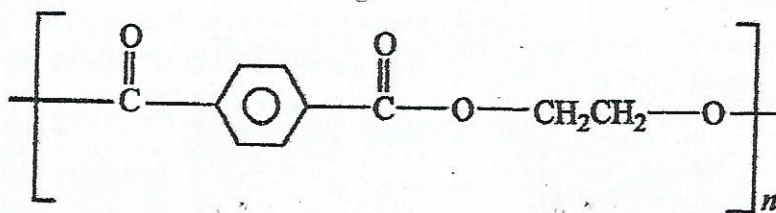


Fig. 8.1

- (i) Draw and name the structures of the monomers that react to form X.
- (ii) Name the type of polymerisation that results in formation of X.
- (iii) With the aid of a chemical equation explain why a fabric made of X is weakened when placed in aqueous NaOH.

[7]

- (b) Describe and explain what happens when

- (i) a mixture of ethanol and acidified sodium dichromate (VI) solution is heated,
- (ii) excess chlorine is added to boiling methylbenzene in the presence of UV light.

[5]
[Total:12]

